

ReWorded.

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COGS 189

Objective.

Our objective is to investigate how **Event Related Potentials (ERPs)**, particularly the N400 component, can be used to facilitate real-time word recognition. We aim to identify the optimal frequency and electrode placement that balances user comfort with strong neural responses for language processing. We intend to determine whether distinct EEG waveform characteristics, specifically those related to the N400, can effectively differentiate between real words and non-words. This comprehensive approach will enhance our understanding of cognitive processing during language comprehension and assess the effectiveness of machine learning models in capturing these differences, ultimately finding the rate at which the brain most effectively processes and distinguishes letters and words with minimal noise.

Introduction and Motivation

Word recognition is a fundamental cognitive process that allows us to rapidly interpret written and spoken language. It engages multiple neural pathways, involving both visual and higher order cognitive areas responsible for meaning and context. One of the most studied neural responses in this domain is N400 component of Event-Related Potentials, which is associated with semantic processing and the brain's ability to detect meaningful versus unexpected words.

Research in this area can aid in diagnosing and treating language-related disorders such as dyslexia and aphasia while also contributing to advancements in artificial intelligence, BCIs, and Neurotechnology. By examining how the brain distinguishes between real words and non-words, we can deepen our understanding of language comprehension and improve assistive technologies for individuals with communication challenges.

Related Literature.

Zaidel, E., Hill, A., & Weems, S. (1970) – similar experiment

Visual word recognition is a multifaceted cognitive task that involves various brain regions and is influenced by the orthographic, phonological, and semantic properties of words, highlighting the synthesis of these processes in understanding language (**Carreiras et al. 2014**).

The EEG is characterized by superior temporal resolution, and recent methods for source localization have enriched the method with improved spatial resolution as well.

Lutzenberger, Pulvermuller, and Birbaumer (1994) showed greater activation in the gamma band (25-30 Hz) for word presentations compared to nonword presentations during lexical decision.

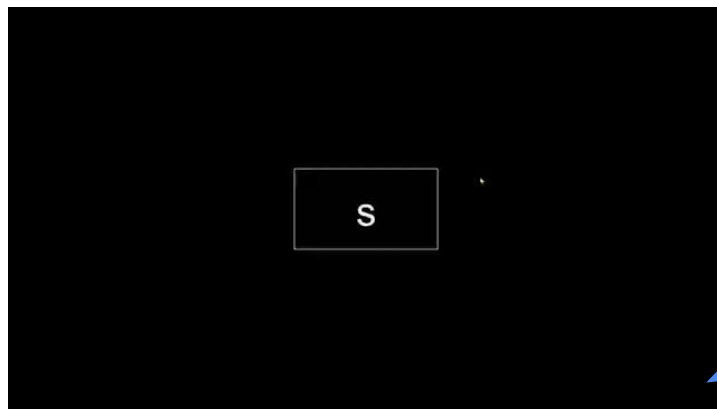
Hypothesis.

Our analysis aims to determine if distinct EEG waveform characteristics can be associated with different types of word predictions, specifically real words versus non-words. Based on the performance of our machine learning models, which exhibited moderate accuracy in classifying these types, we hypothesize that the EEG waveform features utilized do not significantly distinguish between these prediction types. This suggests that the EEG characteristics of real words and non-words may not be as clearly differentiable as anticipated under the conditions of our study.

Experimental Design - Visual Stimuli

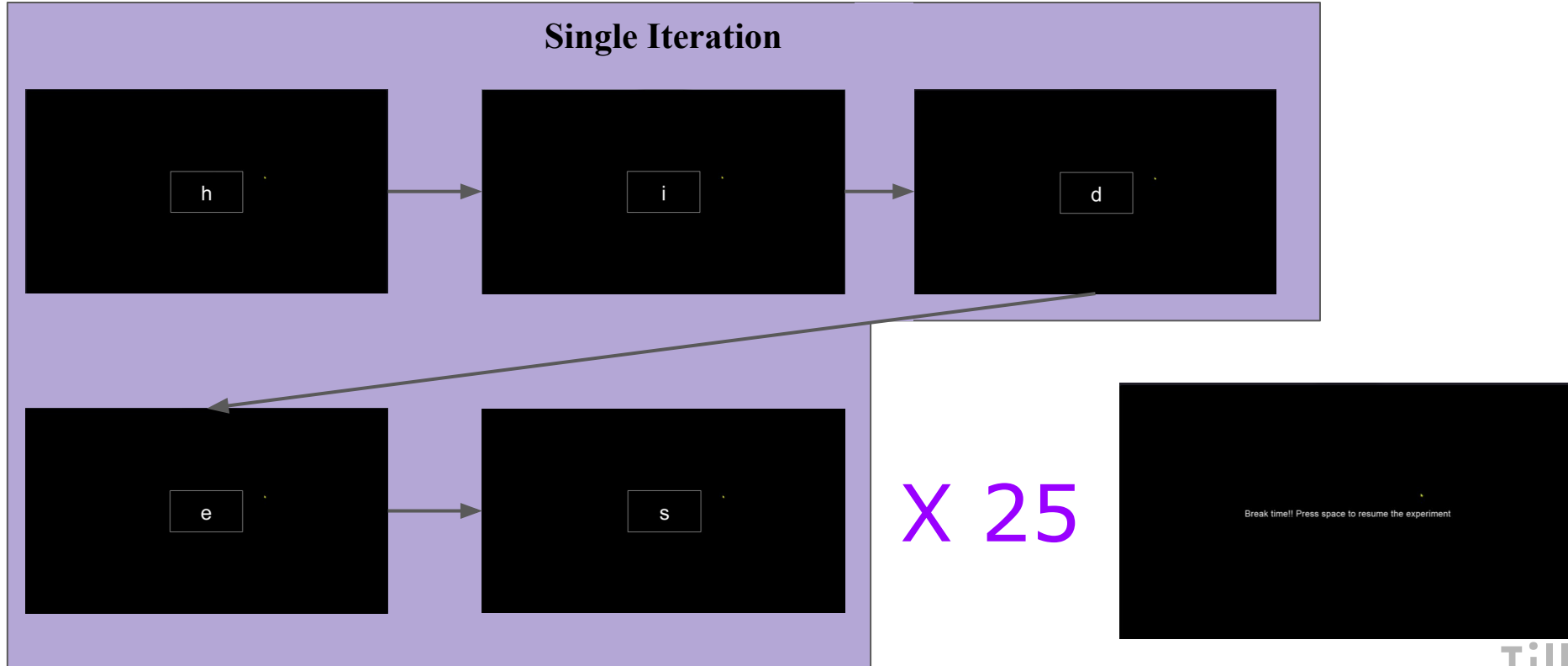
The participants are presented with a string of letters displayed sequentially, with each letter appearing for ~500 millisecond (to minimize eye movements and ensure timing for EEG collection).

The letters are paired with a flicking stimuli on the bottom right corner (15 Hz) to synchronize signals.

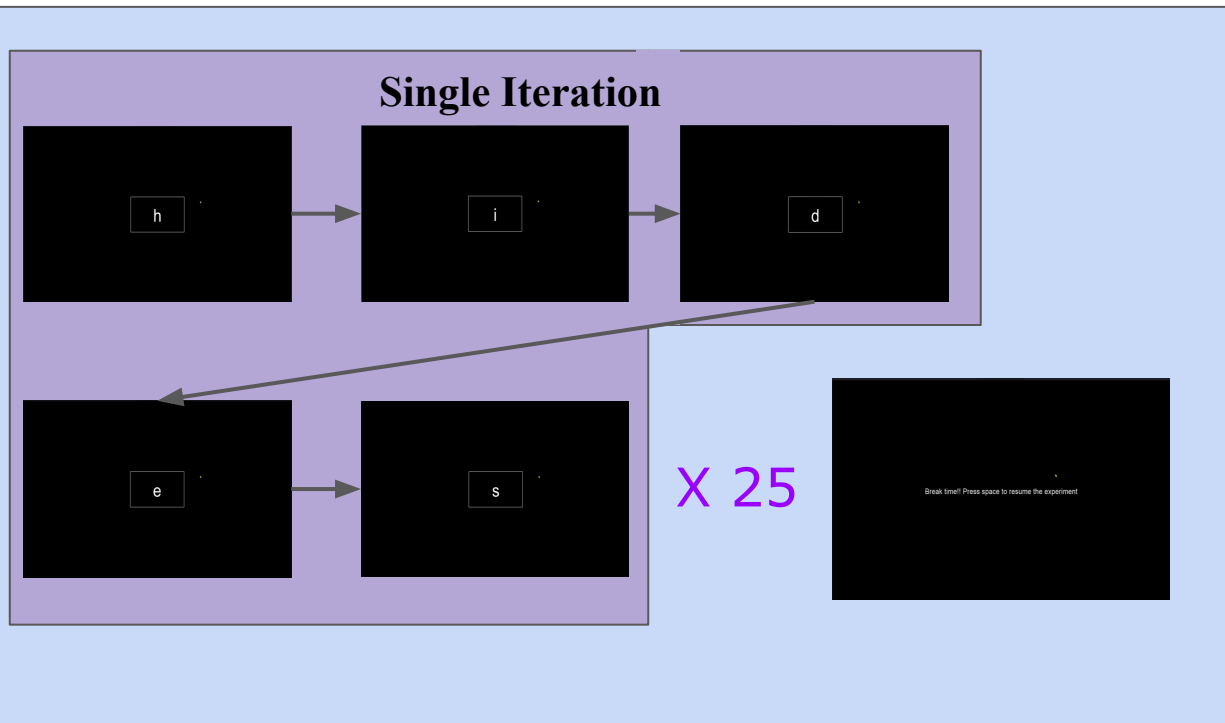


Time Locking
Features

Experimental Design - Display



Experimental Design - Display

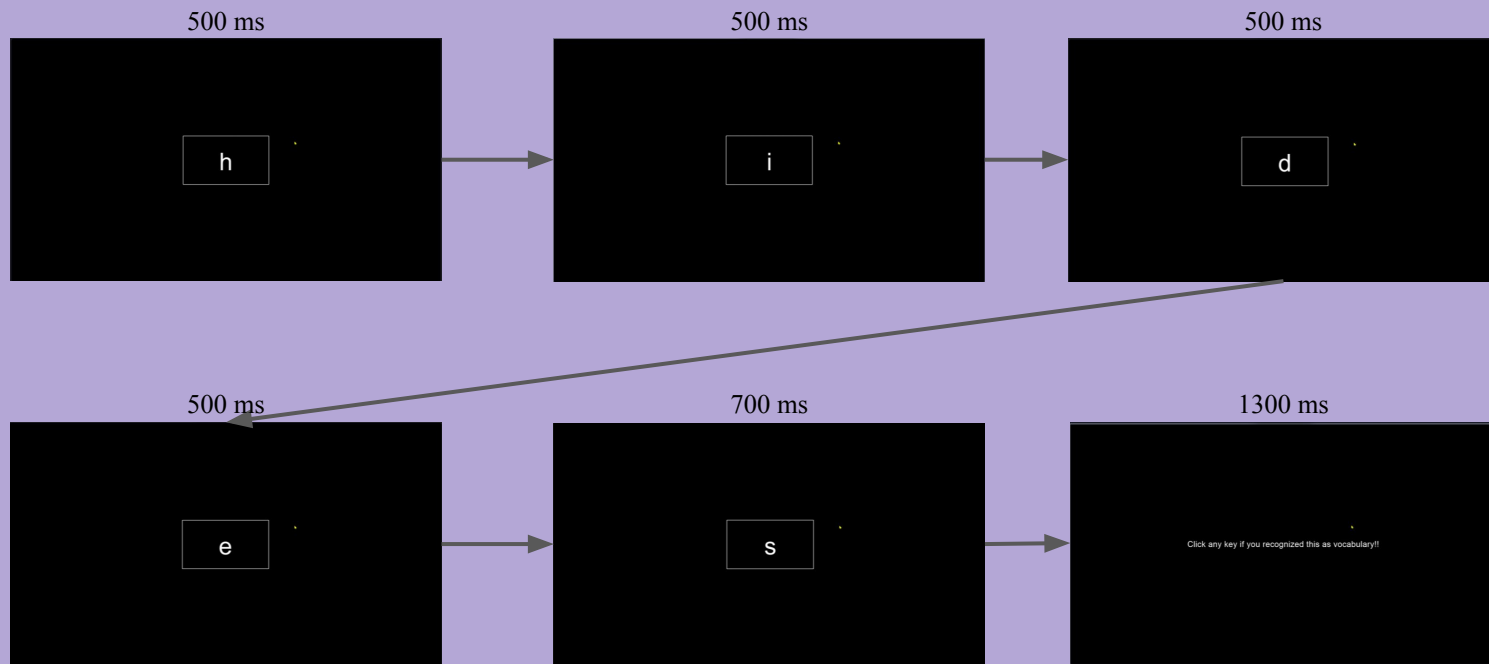


X 8 = 200
iterations

Experimental Design - Display

Jill

Single Iteration - 4 sec

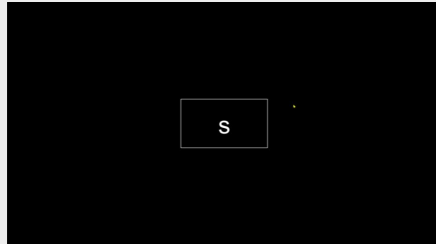


Experimental Design - Display

Single Iteration - 4 sec



Data Collection



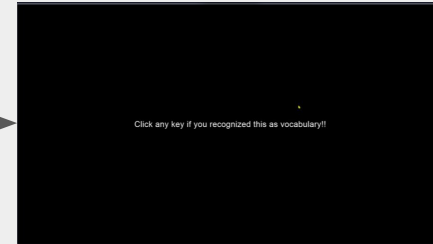
Sample rate: 250Hz

Time: 700ms

Signal as microVolts

Stored in CSV file

User Label



Users response of whether the word was a vocabulary or random combination of characters

Methods - Data

Text Corpus: popular.txt

- List of 25,322 common english word
- Found in: <https://github.com/dolph/dictionary>



Label = 1 data: correct.txt

- Filtered to word of 5 char
- 3088 words

Label = 0 data: incorrect_trick.txt

- Randomly change the last character
- Intended so user can't immediately jump to conclusion
- exp) awful → awfuj

Label = 0 data: incorrect_advnace.txt

- While keeping vowel location, shuffled the constant presented in each word
- Intended to avoid obviously incorrect sequence of word such as “vw”
- exp) bugyr → rugby

Methods - Data

Label = 1 data: correct.txt

- Filtered to word of 5 char
- 3088 words

50%

Label = 0 data: incorrect_trick.txt

- Randomly change the last character
- Intended so user can't immediately jump to conclusion
- exp) awful → awfuj

25%

Label = 0 data: incorrect_advnce.txt

- While keeping vowel location, shuffled the constant presented in each word
- Intended to avoid obviously incorrect sequence of word such as “vw”
- exp) bugyr → rugby

25%

Methods.

Ch0: **Fp1** → left prefrontal cortex (working memory + language comprehension)

Ch1: **F7** → left anterior temporal (semantic memory)

Ch2: **F3** → left dorsolateral prefrontal cortex (working memory)

Ch3: **C3** → left motor cortex (hand movements)

Ch4: **T5** → left visual word form area (word identification)

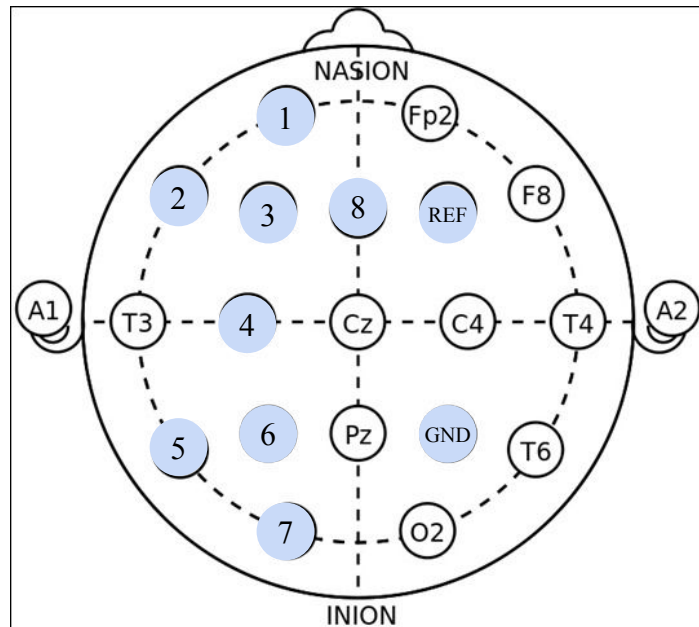
Ch5: **P3** → left wernicke's area (language comprehension)

Ch6: **O1** → left occipital lobe (visual stimuli)

Ch7: **Fz** → midline frontal (memory and attention)

Function:
Cancel out noise

[	REF: F4 → similar to F3, but on the right side
	GND: P4 → right parietal lobe (visual-spatial processing and attention)



We found that majority of the electrode placements that we are measuring is on the left hemisphere, which is dominant for language processing.

Dataset

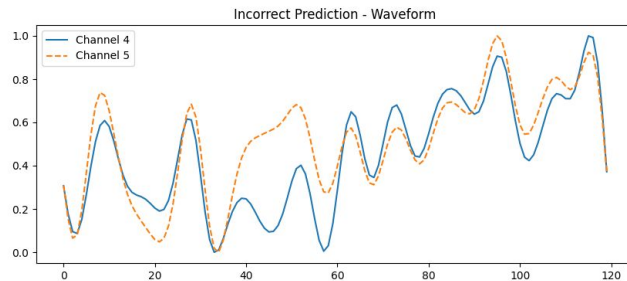
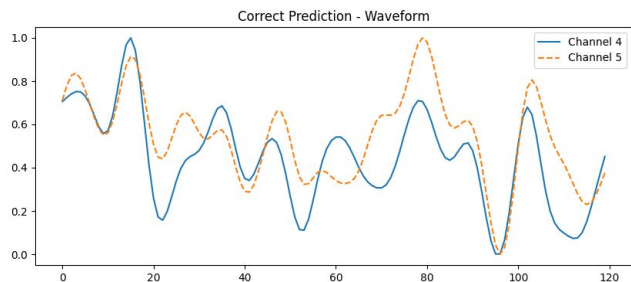
word	label	user_input	channel_0	channel_1	channel_2	channel_3	channel_4	channel_5	channel_6	channel_7
merit	1	1	[90.0, 91.0, 92.0, 93.0, 94.0, 95.0, 96.0, 97....	[11787.99710130657, 11763.812513805928, 11710....	[11254.617423369578, 11231.148091691506, 11158...	[16406.38159589548, 16390.44480209885, 16337.2....	[-17321.90904878486, -17351.1451305324, -17418...	[4174.3223874953255, 4153.6023203852565, 4113....	[1937.5162646193821, 1919.6572207995919, 1870....	[8696.236753015131, 8693.845116358414, 8639.53...
scark	0	0	[148.0, 149.0, 150.0, 151.0, 152.0, 153.0, 154...	[11659.54162592192, 11607.126785174225, 11583....	[11220.866289242065, 11174.933454386408, 11144...	[16427.034607772184, 16376.51966530319, 16355....	[-17294.75167927166, -17359.3482207475, -17382...	[4181.407890487658, 4135.922090521109, 4105.32...	[2006.6725619641022, 1958.8845323186556, 1941....	[8785.062585480522, 8740.82848320347, 8719.661...
trunk	1	1	[86.0, 87.0, 88.0, 89.0, 90.0, 91.0, 92.0, 93....	[11464.969690438473, 11497.714996065497, 11595...	[11088.990996955752, 11124.7761398287, 11208.5...	[16327.502289712704, 16352.402133035916, 16425...	[-17421.709587777805, -17377.0508023561, -1728...	[4017.480196652436, 4043.452923709503, 4113.12...	[1908.0566654272875, 1941.4948751324268, 2013....	[8654.215473439155, 8683.831534842437, 8756.58...

Addressed Issue in Data Collection

- Some of the rows (every two - three iteration) had no data collected by the board
 - No specific patterns in missingness
- Issue: There was a lag between the time row is recorded to when the data from the board is relayed
- Ensured that the data for each channel is extracted before moving onto next iteration

Analysis.

- EEG waveform features used to train the models included average amplitude, peak-to-peak difference, standard deviation, as well as the highest and lowest peak amplitudes. These features were selected to capture the variability and extremities in the EEG signals, which are indicative of neural responses during word recognition tasks.
- We incorporated correlations between EEG channels 4 (T5) and 5 (P3), important for detecting the N400 component in semantic processing, as model features. However, these correlations did not significantly enhance the predictive accuracy for distinguishing between real words and non-words.



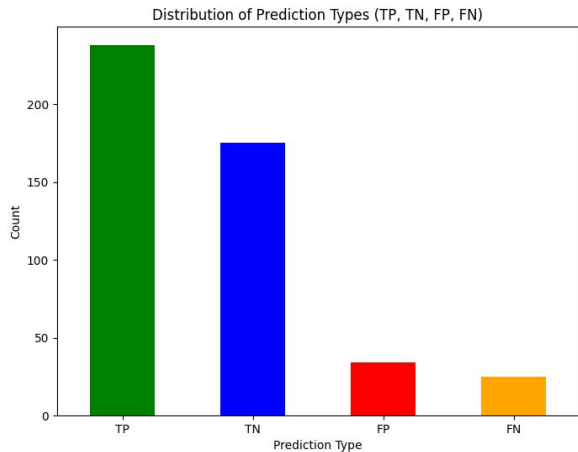
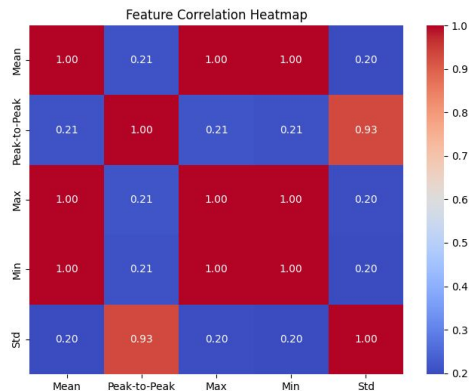
Results.

- The Support Vector Machine (SVM) with a linear kernel achieved an accuracy of approximately 51.58% in predicting the type of word (real word vs. non-word).
- The Gradient Boosting Machine (GBM) and Multilayer Perceptron (MLP) models yielded accuracies of 49.47% and 44.21%, respectively, for the same task.
- User identification accuracies were significantly higher, with the SVM and MLP both achieving around 86.32%, and the GBM reaching 83.16%.

Interpretation.

Marco

- Based on model accuracies, the characteristics of the waveforms were not unique enough to differentiate between true positive and negative and false positive and negative.
- They were unique enough to be able to tell whether the user was able to correctly categorize the word



Limitations

- The limited sample size and potential variability in EEG signal quality may have negatively impacted the performance of the machine learning models.
- Conduct experiments on a larger and more diverse participant pool at multiple time points to better understand and account for variability in the data.
- Feature extraction was focused primarily on waveform characteristics, which may not capture all the cognitive dimensions associated with semantic processing as reflected in the N400 component.

Discussion.

We learned that measuring and interpreting brain processes is complex. There were many times that we hit a roadblock. If we had more time, we would gather more data for our machine learning model to improve accuracy. One of the things we realized was that we needed to confirm everything before data collection. We have data we couldn't use due to difference in conditions (ex. electrode placements).

Improvements

- More data, the better the accuracy!
 - Our machine learning model was trained with 3 datasets that we gathered.
- Diverse participant pool
 - We were the subjects. It would be better if there were outside participants to account for possible variability.

Conclusion.

- Our study, focusing on the N400 component within Event-Related Potentials (ERPs) in EEG data, has significantly enriched our understanding of semantic processing. It reveals that specific EEG waveform characteristics, while insightful, do not distinctly differentiate between real words and non-words as initially hypothesized. This outcome challenges pre-existing assumptions and prompts further exploration into the subtle and complex neural mechanisms that support language comprehension.
- Moving forward, further research could explore additional EEG features, more complex machine learning algorithms, and an expanded dataset to enhance the robustness and accuracy of the findings to accurately capture the nuances of brain activity associated with ERPs during semantic processing. These advancements will not only refine our comprehension of cognitive processes but also enhance the accuracy of technologies aimed at real-time word recognition, supporting broader applications in neurolinguistics and cognitive neuroscience.

References.

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Cap Placement

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- Tomberg, C. (1999). Focal enhancement of chaotic strange attractor dimension in the left semantic (Wernicke) human cortex during reading without concomitant change in vigilance level. *Neuroscience Letters*, 263 (2-3), 177-180.